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Brief paper

An extremum seeking-based approach for Nash equilibrium seeking in N-cluster noncooperative games☆

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ABSTRACT

This paper considers Nash equilibrium seeking for N-cluster noncooperative games in which the explicit expressions of the agents' local objective functions are not available to the agents. As an alternative, the output values of the agents' local objective functions are supposed to be measurable. An extremum seeker is designed to achieve the Nash equilibrium seeking for the N-cluster noncooperative games. The design of the extremum seeker is based on a dynamic average consensus protocol and the modulation of the sinusoidal dither signals. Through Lyapunov stability analysis, the convergence results are analytically studied. Local and nonlocal convergence results are derived under local and global characterizations of the Nash equilibrium for the N-cluster noncooperative games, respectively. Compared with our previous works on Nash equilibrium seeking for the N-cluster noncooperative games, the main advantages and characteristics of the proposed method are threefold. Firstly, the proposed method broadens the applicability of the existing Nash equilibrium seeking strategies to circumstances in which the agents cannot access the explicit expressions of their local objective functions. Moreover, the proposed seeking strategy performs as a unified strategy that solves the noncooperative games and the social cost minimization problems without utilizing explicit model information. Secondly, the proposed seeking strategy requires each agent to update fewer auxiliary variables compared with the existing methods and hence reduces the communication and computation costs. Thirdly, different from most of the existing works that adopt singular perturbation to analyze the stability of extremum seekers, this paper establishes the convergence results via Lyapunov stability analysis, which is novel and provides some new insights on the analysis of extremum seekers. A numerical example is provided to verify the effectiveness of the proposed method.

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1. Introduction

Noncooperative games, which serve as mathematical modeling and analysis tools that explore the competition among individual entities (see, e.g., Frihauf, Krstic, & Basar, 2012; Ma, Ye, Zheng, & Zhu, 2019; Stankovic, Johansson, & Stipanovic, 2012; Ye & Hu, 2017a, 2017b, 2018; Yin, Shanbhag, & Mehta, 2011), have found their wide applications in various fields including economic markets (Frihauf et al., 2012), sensor networks (Stankovic et al., 2012), wireless communication networks (Yin et al., 2011), smart grids (Ye & Hu, 2017b), just to name a few. Conversely, the study of social cost minimization problem/distributed optimization problem, which concerns with a network of agents that collaborate to achieve the global objectives, is based on the cooperation among the agents (see, e.g., Bai, Ye, Sun, & Hu, 2019; Li, Yu, Yu, Huang, & Liu, 2016; Rabbat & Nowak, 2004; Raffard, Tomlin, & Boyd, 2004; Sun, Ye, & Hu, 2017; Ye & Hu, 2016a). This category of problems covers resource allocation problems (Bai et al., 2019; Li et al., 2016), flight formation (Raffard et al., 2004), sensor network applications (Rabbat & Nowak, 2004), and so on. However, in many practical situations, cooperation and competition coexist among the interacting participants. In such situations, outright noncooperative games or distributed optimization techniques may not be sufficient to fully capture both the cooperative and competitive behaviors among the agents. It was

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reported in Peng and Bourne (2009) that resource homogeneity, resource heterogeneity, intensive competition and strategic positioning can lead to the coexistence of cooperation and competition among the engaged participants. Such situations occur in many fields including health-care networks (Peng & Bourne, 2009), transportation networks (Shehory & Kraus, 1998), political science, just to name a few. Driven by the incentive to capture the coexistence of cooperation and competition, an N-cluster noncooperative game was formulated in Ye, Hu, and Lewis (2018) and further studied in Ye, Hu, Lewis, and Xie (2019), where seeking strategies were proposed to solve the formulated game. In particular, Ye et al. (2019) proposed a strategy to reduce the communication and computation costs for the method in Ye et al. (2018). Nevertheless, the proposed seeking strategies in Ye et al. (2018, 2019) were based on the real-time measurements of the gradient flows. This indicates that the explicit expressions of the agents' local objective functions are required to implement the seeking strategies. Motivated by the incentive to broaden the applicability of the N-cluster noncooperative games, this paper intends to propose a new seeking scheme that does not require the explicit expressions of the agents' objective functions. To achieve the objective, the main idea is to design the Nash equilibrium seeking strategy for the formulated N-cluster noncooperative games based on extremum seeking control.

Extremum seeking control, which stands for a class of non-model-based online optimization algorithms, has been widely studied from different perspectives in the past two decades (see, e.g., Durr, Stankovic, Ebenbauer, & Johansson, 2013; Khong, Nesic, Tan, & Manzie, 2013; Krstic & Wang, 2000; Ye & Hu, 2016b; Zhang & Ordonez, 2009). In particular, it has been leveraged to develop Nash equilibrium seeking strategies for noncooperative games (see, e.g., Frihauf et al., 2012; Kutadinata, Moase, & Manzie, 2015; Liu & Krstic, 2011; Stankovic et al., 2012) and distributed optimization/social cost minimization algorithms (see, e.g., Dougherty & Guay, 2017; Kvaternik & Pavel, 2012; Menon & Baras, 2014; Poveda & Quijano, 2013; Vandermeulen, Guay, & McLellan, 2018; Xu & Soh, 2016; Ye & Hu, 2016a). An extremum seeker was designed in Frihauf et al. (2012) to achieve Nash equilibrium seeking among self-interested players. The convergence analysis therein relied on local averaging analysis and singular perturbation analysis, which resulted in local convergence to a neighborhood of the Nash equilibrium. Nonlocal convergence results were investigated in Kutadinata et al. (2015) and a stochastic counterpart was studied in Liu and Krstic (2011). Moreover, a discrete-time stochastic extremum seeking scheme was applied to mobile sensor networks in Stankovic et al. (2012) to solve coordination problems. Recently, researchers have shown a thriving interest towards collaborative extremum seeking for distributed optimization/social cost minimization problems. In Dougherty and Guay (2017), Menon and Baras (2014) and Vandermeulen et al. (2018), collaborative extremum seeking schemes were designed by utilizing a dynamic average consensus protocol. In Ye and Hu (2016a), an extremum seeker was designed based on the primal–dual dynamics to solve distributed optimization problems. Collaborative optimization among a network of dynamic agents was addressed in Kvaternik and Pavel (2012) and resource allocation was considered in Poveda and Quijano (2013).

This paper intends to develop an extremum seeker, that does not require the explicit expressions of the agents' local objective functions, to achieve Nash equilibrium seeking for the N-cluster noncooperative games formulated in Ye et al. (2018). As mentioned in Ye et al. (2018), the clusters behave as self-interested "virtual players" in a noncooperative game while the agents in the same cluster coordinate to minimize their total cost. Hence, the proposed extremum seeker needs to achieve social cost minimization among the agents in the same cluster, while handling their competitive interactions with the agents in other clusters. Compared with the existing works, the main contributions of the paper are summarized as follows.

(1) An extremum seeking-based approach is proposed to seek the Nash equilibrium of the N-cluster noncooperative games. The proposed method adopts a dynamic average consensus protocol to estimate the averaged output values of the clusters' objective functions. The estimated values are then incorporated with sinusoidal dither signals for the update of the agents' actions. Compared with the model-based seeking strategies in Ye et al. (2018, 2019), the proposed method does not require the explicit expressions of the agents' local objective functions to be implemented. Moreover, the proposed method requires each agent to update fewer auxiliary variables compared with the methods in Ye et al. (2018, 2019).
(2) Through Lyapunov stability analysis, the convergence results are derived under certain conditions. With global characterizations on the Nash equilibrium, a nonlocal convergence result is derived. Moreover, local convergence to a neighborhood of each isolated Nash equilibrium that satisfies certain conditions is derived with local characterizations on the Nash equilibria.
(3) The considered problem covers the Nash equilibrium seeking problem studied in Frihauf et al. (2012) and Kutadinata et al. (2015) and the social cost minimization problem studied in Menon and Baras (2014) as special cases. The presented results show that the conditions on the dither frequencies in Frihauf et al. (2012) and Menon and Baras (2014) can be relaxed to some extent. In addition, the presented results improve the local convergence result given in Menon and Baras (2014) to a nonlocal counterpart under certain conditions. Moreover, it is worth noting that unlike the existing literature that adopt singular perturbation to analyze the stability of extremum seekers, this paper conducts Lyapunov stability analysis for the closed-loop system, which provides new insights on the analysis of extremum seekers and improves the convergence results to some extent.

Notations. In this paper, $\mathbb{R}$ is the set of real numbers. The transpose of $\mathbf{x}$ is denoted as $\mathbf{x}^T$. An N-dimensional column vector composite of 1(0) is denoted as $\mathbf{1}_N(\mathbf{0}_N)$ and $\mathrm{diag}\{h_{ij}\}(\mathrm{diag}\{h_i\})$ for $i \in \{1, 2, \ldots, N\}, j \in \{1, 2, \ldots, m_i\}$, where N and m_i are positive integers, is a diagonal matrix whose diagonal elements are $h_{11}, h_{12}, \ldots, h_{1m_1}, h_{21}, \ldots, h_{Nm_N}$ $(h_1, h_2, \ldots, h_N)$, successively. Moreover, $[h_{ij}]_{vec}$ is a column vector defined as $[h_{11}, h_{12}, \ldots, h_{1m_1}, h_{21}, \ldots, h_{Nm_N}]^T$. In addition, $\mathbf{B}_r$ denotes a ball centered at the origin and of radius r. Let $Q \in \mathbb{R}^N \times \mathbb{R}^N$ be a symmetric matrix, then, $\lambda_{min}(Q), \lambda_{max}(Q)$ represent for the smallest and largest eigenvalues of Q, respectively. The smallest value of h_{ij}, where $h_{ij} \in \mathbb{R}$, is represented by $\min\{h_{ij}\}$.

2. Problem formulation

This paper considers the N-cluster noncooperative game formulated in Ye et al. (2018). For the convenience of the readers, the following definition is given to summarize the N-cluster noncooperative game.

Definition 1 (*N-cluster Noncooperative Game*)**.** The N-cluster noncooperative game Γ is defined as a triple $\Gamma \triangleq \{\mathcal{N}, X, f\}$, where $\mathcal{N} = \{1, 2, \ldots, N\}, N \geq 1$[1] is the set of N **virtual** players

[1] Usually, a game would have at least two players. In this formulation, we allow $N = 1$ to cover the social cost minimization problem as a special case of the formulated game.

(termed as "clusters" for clarity), each of which is composite of several decision-making agents. Moreover, $X = X_1 \times \cdots \times X_N$, and $X_i \subseteq \mathbb{R}^{m_i} (X_i = \mathbb{R}^{m_i}$ in this paper) is the set of actions for cluster i governed by **the agents in cluster** i, $m_i \geq 1$ is the number of agents in cluster i. In addition, $f = (f_1, f_2, \ldots, f_N)$, where f_i is the cost function of cluster i defined as

$$f_i(\mathbf{x}_i, \mathbf{x}_{-i}) = \sum_{j=1}^{m_i} f_{ij}(\mathbf{x}_i, \mathbf{x}_{-i}), \forall i \in \mathcal{N}. \tag{1}$$

Note that in (1), $f_{ij}(\mathbf{x}_i, \mathbf{x}_{-i})$ is the local objective function of agent j in cluster i. Moreover, the vector containing the actions of the agents is defined as $\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{im_i}]^T \in X_i$, $\mathbf{x}_{-i} = [\mathbf{x}_1^T, \mathbf{x}_2^T, \ldots, \mathbf{x}_{i-1}^T, \mathbf{x}_{i+1}^T, \ldots, \mathbf{x}_N^T]^T$, $\mathbf{x} = [\mathbf{x}_1^T, \mathbf{x}_2^T, \ldots, \mathbf{x}_N^T]^T$, where $x_{ij} \in \mathbb{R}$ is the action of agent j in cluster i. Without further clarification, $f_i(\mathbf{x}_i, \mathbf{x}_{-i})$ and $f_{ij}(\mathbf{x}_i, \mathbf{x}_{-i})$ might be alternatively written as $f_i(\mathbf{x})$ and $f_{ij}(\mathbf{x})$, respectively, in the rest of the paper.

The problem setting of this paper is given below.

Problem 1. Suppose that the agents in cluster i can communicate with each other via a communication graph $\mathcal{G}_C^i, \forall i \in \mathcal{N}$, and the game admits a Nash equilibrium $\mathbf{x}^*$. The objective of the paper is to develop a Nash equilibrium seeking strategy for game Γ given that the agents can only measure the output values of their own local objective functions.

Note that an action profile $\mathbf{x}^* = (\mathbf{x}_i^*, \mathbf{x}_{-i}^*)$ is said to be the Nash equilibrium of the game if (Ye et al., 2019)

$$f_i(\mathbf{x}_i^*, \mathbf{x}_{-i}^*) \leq f_i(\mathbf{x}_i, \mathbf{x}_{-i}^*), \forall \mathbf{x}_i \in \mathbb{R}^{m_i}, i \in \mathcal{N}. \tag{2}$$

Moreover, the graph related definitions follow those in Ye and Hu (2017a) and are not presented here due to space limitation.

Remark 1. In this paper, only the output values of the agents' local objective functions are supposed to be available in the design of the Nash equilibrium seeking strategy while in our previous works (Ye et al., 2018, 2019), the gradients of the agents' local objective functions were utilized.

Remark 2. Great efforts have been made by researchers to explore the existence, uniqueness and characterizations of Nash equilibrium (see, e.g., Basar & Olsder, 1999; Facchinei & Kanzow, 2007; Facchinei & Pang, 2003; Nash, 1951; Ratliff, Burden, & Sastry, 2016; Rosen, 1965, just to name a few). Though many existing works suppose that the players' actions belong to a convex and compact (non-empty) set for the study of the aforementioned problems, Corollaries 2.2.6–2.2.8 and Theorem 2.3.3 in Facchinei and Pang (2003) provided conditions to ensure the existence or uniqueness of Nash equilibrium for games where the players' actions belong to an unbounded set. Note that the paper focuses on Nash equilibrium seeking without explicit model information and these issues are not discussed in detail to avoid any distraction of the readers' attention. Hence, we directly refer the readers to Basar and Olsder (1999), Facchinei and Kanzow (2007), Facchinei and Pang (2003), Nash (1951), Ratliff et al. (2016) and Rosen (1965) for more detailed discussions on these issues.

The following two assumptions will be utilized throughout the rest of the paper.

Assumption 1. For each $i \in \mathcal{N}$, the communication graph $\mathcal{G}_C^i$ is undirected and connected.

Assumption 2. For each $i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}, f_{ij}(\mathbf{x})$ is smooth.

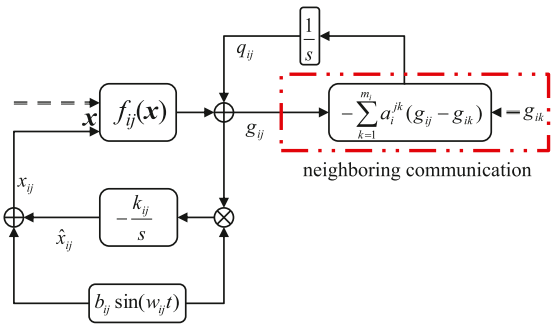


Fig. 1. Illustration of the seeking strategy in (3).

3. Main results

In this section, a new Nash equilibrium seeking strategy will be proposed.

3.1. Nash equilibrium seeking strategy design

Motivated by Ye et al. (2018, 2019), the action updating strategy for agent j in cluster i, where $i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$, is designed as

$$\begin{aligned} &x_{ij} = \hat{x}_{ij} + b_{ij}\sin(w_{ij}t), \dot{\hat{x}}_{ij} = -k_{ij}g_{ij}b_{ij}\sin(w_{ij}t) \\ &g_{ij} = q_{ij} + f_{ij}(\mathbf{x}), \dot{q}_{ij} = -\sum_{k=1}^{m_i} a_i^{jk}(g_{ij} - g_{ik}), \end{aligned} \tag{3}$$

where $b_{ij} = b\bar{b}_{ij}, k_{ij} = \epsilon\delta\bar{k}_{ij}, w_{ij} = \delta\bar{w}_{ij}$ in which b, δ, ϵ are small positive parameters, $\bar{b}_{ij}, \bar{k}_{ij}$ are fixed positive constants, and $\bar{w}_{ij}$ is a fixed positive rational number. Moreover, a_i^{jk} denotes the element on the jth row and kth column of the adjacency matrix of $\mathcal{G}_C^i$. In (3), $\hat{x}_{ij}$ is the estimated action of agent j in cluster i, $q_{ij}, g_{ij}, \forall i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$ are auxiliary variables. In addition, $q_{ij}(t_0) = 0, \forall i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$, where $t_0 \geq 0$ is the initial time instant. The seeking strategy is illustrated in Fig. 1.

3.2. Insights into the strategy design

In the following, we provide some insights on the design of the seeking strategy in (3).

For notational convenience, define $q_i = [q_{i1}, q_{i2}, \ldots, q_{im_i}]^T$, $g_i = [g_{i1}, g_{i2}, \ldots, g_{im_i}]^T$, $\mathcal{F}_i(\mathbf{x}) = [f_{i1}(\mathbf{x}), f_{i2}(\mathbf{x}), \ldots, f_{im_i}(\mathbf{x})]^T \in \mathbb{R}^{m_i}$, $\hat{\mathbf{x}}_i = [\hat{x}_{i1}, \hat{x}_{i2}, \ldots, \hat{x}_{im_i}]^T$, $\hat{\mathbf{x}} = [\hat{x}_{ij}]_{vec}$, $\bar{\mathbf{x}} = [\bar{x}_{ij}]_{vec} = \hat{\mathbf{x}} - \mathbf{x}^*$.

To extract the uncontrollable variable from (3), let $R_i \in \mathbb{R}^{m_i} \times \mathbb{R}^{m_i - 1}$ be a matrix such that $U_i = \left[\mathbf{1}_{m_i}/\sqrt{m_i} \;\; R_i\right]$ is an orthogonal matrix. Moreover, let $[\underline{g}_i^T, \bar{g}_i^T]^T = U_i^T g_i$, $[\underline{q}_i^T, \bar{q}_i^T]^T = U_i^T q_i$, where $\underline{g}_i, \underline{q}_i \in \mathbb{R}, \bar{g}_i, \bar{q}_i \in \mathbb{R}^{m_i - 1}$ (Ye & Hu, 2017b). Denote the Laplacian matrix of $\mathcal{G}_C^i$ as $\mathcal{L}_i, i \in \mathcal{N}$. Then, under Assumption 1, the system in (3) can be transformed into

$$\begin{aligned} &\dot{\underline{q}}_i = 0, \dot{\bar{q}}_i = -R_i^T\mathcal{L}_i R_i \bar{g}_i \\ &\underline{g}_i = \underline{q}_i + \frac{\mathbf{1}_{m_i}^T}{\sqrt{m_i}}\mathcal{F}_i(\mathbf{x}), \bar{g}_i = \bar{q}_i + R_i^T\mathcal{F}_i(\mathbf{x}) \\ &\dot{\hat{x}}_{ij} = -k_{ij}g_{ij}b_{ij}\sin(w_{ij}t), x_{ij} = \hat{x}_{ij} + b_{ij}\sin(w_{ij}t), \end{aligned} \tag{4}$$

$i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$, by which it can be derived that $\underline{q}_i(t) = \underline{q}_i(t_0) = 0, \forall i \in \mathcal{N}, t \geq t_0$.

Let $\tau = \delta t$. Then, in the τ-time scale

$$\begin{aligned}&\delta\frac{d\bar{q}_i}{d\tau} = -R_i^T\mathcal{L}_iR_i\bar{g}_i,\ \frac{d\hat{x}_{ij}}{d\tau} = -\epsilon\bar{k}_{ij}g_{ij}b_{ij}\sin(\bar{w}_{ij}\tau)\\&\underline{g}_i = \underline{q}_i + \frac{\mathbf{1}_{m_i}^T}{\sqrt{m_i}}\mathcal{F}_i(\mathbf{x}),\ \bar{g}_i = \bar{q}_i + R_i^T\mathcal{F}_i(\mathbf{x}),\end{aligned} \tag{5}$$

by which it can be derived that

$$\begin{aligned}\delta\frac{d\bar{g}_i}{d\tau} =&\delta\frac{d\bar{q}_i}{d\tau} + \delta R_i^T\frac{d\mathcal{F}_i(\mathbf{x})}{d\tau}\\ =&-R_i^T\mathcal{L}_iR_i\bar{g}_i + \delta R_i^T\left(\frac{\partial\mathcal{F}_i(\mathbf{x})}{\partial\mathbf{x}}\right)^T\frac{d\mathbf{x}}{d\tau}.\end{aligned} \tag{6}$$

Letting $\delta = 0$ "freezes" $\bar{g}_i = \mathbf{0}_{m_i-1}$, on which $g_i = \frac{\mathbf{1}_{m_i}}{\sqrt{m_i}}\underline{g}_i + R_i\bar{g}_i = \frac{\mathbf{1}_{m_i}}{m_i}f_i(\mathbf{x})$, thus giving the **reduced-system** as

$$\frac{d\hat{x}_{ij}}{d\tau} = -b\epsilon\frac{\bar{b}_{ij}\bar{k}_{ij}}{m_i}f_i(\mathbf{x})\sin(\bar{w}_{ij}\tau), \tag{7}$$

$\forall i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$.

Moreover, define $t' = \frac{\tau-\tau_0}{\delta}$. Then, for $i \in \mathcal{N}$,

$$\begin{aligned}&\frac{d\bar{g}_i}{dt'} = \delta\frac{d\bar{q}_i}{d\tau} + \delta R_i^T\left(\frac{\partial\mathcal{F}_i(\mathbf{x})}{\partial\mathbf{x}}\right)^T\frac{d\mathbf{x}}{d\tau} = -R_i^T\mathcal{L}_iR_i\bar{g}_i\\&+\delta R_i^T\left(\frac{\partial\mathcal{F}_i(\mathbf{x})}{\partial\mathbf{x}}\right)^T\left(\frac{d\hat{\mathbf{x}}}{d\tau} + [b_{ij}\bar{w}_{ij}\cos(\bar{w}_{ij}\tau)]_{vec}\right).\end{aligned} \tag{8}$$

Letting $\delta = 0$ gives the **boundary-layer system** (see Chapter 11 of Khalil, 2002) as

$$\frac{d\bar{g}_i}{dt'} = -R_i^T\mathcal{L}_iR_i\bar{g}_i,\ i \in \mathcal{N}. \tag{9}$$

Remark 3. From the above analysis, it can be seen that the design of the Nash equilibrium seeking strategy in (3) is based on (7). As each agent can only measure the output value of its own local objective function and the cluster's objective function is not available to the agents, (7) cannot be directly implemented. To overcome the unavailability of the clusters' objective functions, a dynamic average consensus protocol is employed in (3) for information dissemination. Moreover, in (7), the sinusoidal dither signals are utilized to extract the required gradient values.

3.3. Convergence analysis

In the following, a nonlocal convergence result will be firstly studied, followed by a local convergence result.

3.3.1. Global characterizations of Nash equilibrium

The nonlocal convergence result developed in the paper is based on the following assumption.

Assumption 3. For all distinct $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{\sum_{i=1}^N m_i}$,

$$(\mathbf{x}-\mathbf{y})^T(\mathcal{P}(\mathbf{x}) - \mathcal{P}(\mathbf{y})) > 0, \tag{10}$$

where $\mathcal{P}(\mathbf{x}) = [(\frac{\partial f_1(\mathbf{x})}{\partial\mathbf{x}_1})^T, (\frac{\partial f_2(\mathbf{x})}{\partial\mathbf{x}_2})^T, \ldots, (\frac{\partial f_N(\mathbf{x})}{\partial\mathbf{x}_N})^T]^T$.

Remark 4. It is worth mentioning that Assumption 3 is a commonly adopted assumption (see, e.g., Rosen, 1965; Ye & Hu, 2017a; Ye et al., 2019; Zhu & Frazzoli, 2016) that globally characterizes a Nash equilibrium in the considered game and ensures the uniqueness of the Nash equilibrium. Moreover, by Assumption 3, $\mathbf{x}$ is at the Nash equilibrium if and only if $\mathcal{P}(\mathbf{x}) = \mathbf{0}$.

Define $\bar{\mathbf{g}} = [\bar{g}_1^T, \bar{g}_2^T, \ldots, \bar{g}_N^T]^T$ and $\mathcal{E}_r(t) = [\bar{\mathbf{x}}^T(t), \bar{\mathbf{g}}^T(t)]^T$. Then, by utilizing $\mathcal{K}$ and $\mathcal{KL}$ functions (see Definitions 4.2–4.3 of Khalil, 2002), the nonlocal convergence result can be described as follows.

Theorem 1. *Suppose that Assumptions 1–3 are satisfied and $\bar{w}_{ij}$ for all $i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$ are distinct from each other. Then, for any pair of strictly positive real numbers (Δ, v), there exist $\epsilon^*(\Delta, v) > 0, b^*(\Delta, v) > 0$ such that for each fixed $\epsilon \in (0, \epsilon^*)$, $b \in (0, b^*)$, there exists $\delta^*(\Delta, v, \epsilon, b) > 0$ such that for each $\delta \in (0, \delta^*)$, there exists a function $\beta \in \mathcal{KL}$ such that*

$$\|\mathcal{E}_r(t)\| \le \beta(\|\mathcal{E}_r(t_0)\|, \epsilon b^2\delta(t-t_0)) + v, \tag{11}$$

for all $t \ge t_0 \ge 0$ with $\|\mathcal{E}_r(t_0)\| \le \Delta$.

Proof. See Appendix A for the proof. □

Remark 5. Note that Theorem 1 presents a semi-global convergence result rather than a local result as the initial error is not required to be sufficiently small. Instead, it assumes that the initial error is bounded and the initial error bound can be arbitrarily large as in the statement of Theorem 1, Δ can be any positive real number. In addition, as v can be any positive real number, Theorem 1 suggests that the proposed method can drive the agents' actions to arbitrarily small neighborhood of the Nash equilibrium.

Theorem 1 is established under Assumption 3, by which the Nash equilibrium is unique. In Frihauf et al. (2012), the authors provided local convergence results for N-**player** noncooperative games under milder conditions. In the following, we show that under the local characterization conditions in Frihauf et al. (2012), compatible convergence results can be obtained for the considered N-**cluster** noncooperative games by utilizing the proposed method.

3.3.2. Local characterizations of Nash equilibrium

To characterize a **local** Nash equilibrium (Ratliff et al., 2016), we adapt the conditions in Frihauf et al. (2012) as follows.

Assumption 4 (*Ye & Hu, 2017a; Ye et al., 2018*)**.** There exists at least one, **possibly multiple** isolated Nash equilibria $\mathbf{x}^*$ on which $\frac{\partial f_i}{\partial\mathbf{x}_i}(\mathbf{x}^*) = \mathbf{0}_{m_i}$, $\forall i \in \mathcal{N}$, and for each $i \in \mathcal{N}$, $\frac{\partial^2 f_i}{\partial\mathbf{x}_i^2}(\mathbf{x}^*)$ is symmetric positive definite.

Assumption 5 (*Ye & Hu, 2017a; Ye et al., 2018*)**.** The matrix

$$B = \begin{bmatrix}\frac{\partial^2 f_1}{\partial\mathbf{x}_1^2}(\mathbf{x}^*) & \frac{\partial^2 f_2}{\partial\mathbf{x}_2\partial\mathbf{x}_1}(\mathbf{x}^*) & \cdots & \frac{\partial^2 f_N}{\partial\mathbf{x}_N\partial\mathbf{x}_1}(\mathbf{x}^*)\\ \frac{\partial^2 f_1}{\partial\mathbf{x}_1\partial\mathbf{x}_2}(\mathbf{x}^*) & \frac{\partial^2 f_2}{\partial\mathbf{x}_2^2}(\mathbf{x}^*) & & \vdots\\ \vdots & & \ddots & \\ \frac{\partial^2 f_1}{\partial\mathbf{x}_1\partial\mathbf{x}_N}(\mathbf{x}^*) & \cdots & & \frac{\partial^2 f_N}{\partial\mathbf{x}_N^2}(\mathbf{x}^*)\end{bmatrix}^T,$$

where $\frac{\partial^2 f_i(\mathbf{x})}{\partial\mathbf{x}_i\partial\mathbf{x}_j} = \begin{bmatrix}\frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i1}\partial x_{j1}} & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i1}\partial x_{j2}} & \cdots & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i1}\partial x_{jm_j}}\\ \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i2}\partial x_{j1}} & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i2}\partial x_{j2}} & \cdots & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{i2}\partial x_{jm_j}}\\ \vdots & & \ddots & \vdots\\ \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{im_i}\partial x_{j1}} & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{im_i}\partial x_{j2}} & \cdots & \frac{\partial^2 f_i(\mathbf{x})}{\partial x_{im_i}\partial x_{jm_j}}\end{bmatrix}^T$, is strictly diagonally dominant.

Remark 6. Assumption 4 is motivated by the results on nonlinear programming, where the stationary condition ensures a critical point and the second-order condition ensures that the critical point is a local minimum point. Assumption 5 is utilized to ensure that the Nash equilibrium is stable under the gradient play governed by

$$\dot{x}_{ij} = -\bar{k}_{ij}\frac{\partial f_i(\mathbf{x})}{\partial x_{ij}}, \tag{12}$$

where $\bar{k}_{ij}$ is a positive control gain. By linearizing (12), it can be seen that Assumptions 4–5 are almost the mildest conditions that are independent of the control gains to ensure the convergence of (12). Though, it might be challenging to check Assumptions 4–5 directly, one may check some more conservative conditions, which may provide evidences to show that Assumptions 4–5 are satisfied. For example, for quadratic games, Assumption 3.1 in Frihauf et al. (2012) can be checked instead. Moreover, for potential games, it would be sufficient to check the strong convexity of the potential function to obtain the corresponding convergence result (see Corollary 3 in Ye & Hu, 2017a), just to name a few.

Under Assumptions 4–5, the following result can be derived.

Theorem 2. *Suppose that Assumptions 1–2, 4–5 are satisfied and $\bar{w}_{ij}$ for all $i \in \mathcal{N}, j \in \{1, 2, \ldots, m_i\}$ are distinct from each other. Then, for any strictly positive real number v, there exist $\epsilon^*(v) > 0, b^*(v) > 0$ such that for each fixed $\epsilon \in (0, \epsilon^*)$, $b \in (0, b^*)$, there exists a positive constant $\delta^*(\epsilon, b)$ such that for each $\delta \in (0, \delta^*)$, there exists a positive constant $\mathcal{T}$ such that*

$$\begin{aligned} \|\mathcal{E}_r(t)\| \leq & \lambda_1 e^{-\delta\epsilon b^2\lambda_2(t-t_0)} + O(\epsilon b), \forall t_0 \leq t \leq t_0 + \mathcal{T} \\ \|\mathcal{E}_r(t)\| \leq & v + O(\epsilon b), \forall t \geq t_0 + \mathcal{T}, \end{aligned} \tag{13}$$

for some positive constants λ_1, λ_2, given that $\|\mathcal{E}_r(t_0)\|$ is sufficiently small.

Proof. See Appendix B for the proof. □

Remark 7. Theorems 1–2 provide some hints on the tuning of the parameters including ϵ, b and δ. From the proofs of the theorems, we see that the existences of $\epsilon^*, b^*, \delta^*$ (stated in the theorems) are ensured. However, as the explicit expressions of the agents' objective functions are unknown, it is in general not possible to analytically determine the precise values of ϵ^*, b^* and δ^*. Nevertheless, we believe that the presented results are useful as the assumptions utilized in the paper are quite mild to include a large class of games and we only need to choose ϵ, b, δ to be sufficiently small in practice.

Remark 8. Note that unlike most of the existing works on perturbation-based extremum seekers, where the convergence analyses were conducted via singular perturbation, this paper conducts the convergence analysis for the closed-loop system by direct Lyapunov stability analysis, which is novel in the extremum seeking literature. Though Lemma 1 in Tan, Nesic, and Mareels (2006) might be utilized to establish the convergence results, it would result in a stability result that is uniformly[2] in (ϵ, b^2) but not in δ. In contrast, the direct Lyapunov stability analysis in this paper gives better convergence result in the sense that the "uniformity in δ" is proven. Moreover, when $m_i = 1, \forall i \in \mathcal{N}, N \geq 2$, the formulated N-cluster noncooperative game is reduced to an N-player noncooperative game studied in Frihauf et al. (2012) and when $N = 1, m_1 \geq 2$, the formulated N-cluster game is reduced to a social cost minimization problem studied in Menon and Baras (2014). It is worth noting that the presented results only require that the dither frequencies are distinct from each other while the results in Frihauf et al. (2012) and Menon and Baras (2014) required that $\bar{w}_{i1} \neq \bar{w}_{j1}$, $2\bar{w}_{i1} \neq \bar{w}_{j1}$ and $2\bar{w}_{i1} = \bar{w}_{j1} + \bar{w}_{k1}$ for all distinct $i, j, k \in \mathcal{N}$. Hence, the presented convergence results further relax the conditions on the dither frequencies in Frihauf et al. (2012) and Menon and Baras (2014) to some extent. Moreover, for $N = 1, m_1 \geq 2$, Theorem 1 improves Menon and Baras (2014) to establish a nonlocal convergence result.

[2] Note that the "uniform" utilized here follows the definitions given in Tan et al. (2006).

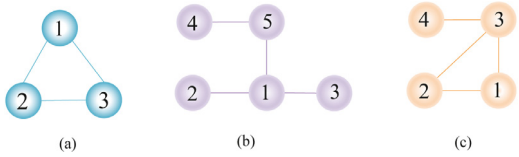


Fig. 2. The communication graphs for the agents in clusters 1–3 are given in (a)–(c), respectively.

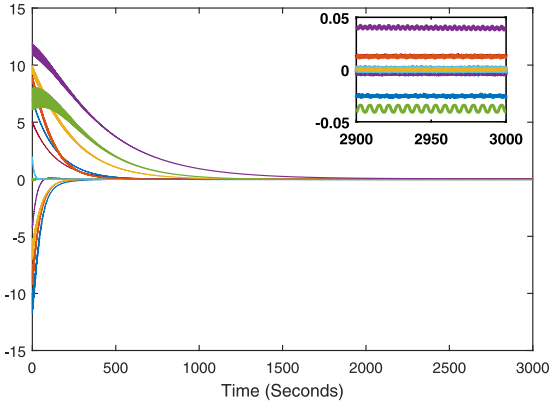


Fig. 3. The agents' estimated actions $\hat{x}_{ij}(t), i \in \{1, 2, 3\}, j \in \{1, 2, \ldots, m_i\}$ generated by the proposed method in (3).

4. A numerical example

In this section, we verify the analytical results in Section 3 via conducting numerical simulations.

Example 1. Consider a 3-cluster noncooperative game, in which each cluster has 3, 5 and 4 agents ($m_1 = 3, m_2 = 5, m_3 = 4$), respectively. Let γ_1 be a constant to be determined and let the objective functions of the agents in cluster 1 be $f_{11}(\mathbf{x}) = 5x_{11}^2 - \gamma_1 x_{11}^3 + (x_{11}x_{21} + x_{11}x_{31}), f_{12}(\mathbf{x}) = \frac{1}{2}(-x_{11}x_{12} + 4x_{12}^2 + 3x_{13}^2), f_{13}(\mathbf{x}) = f_{12}(\mathbf{x})$, respectively. Moreover, the objective functions for the agents in cluster 2 are $f_{21}(\mathbf{x}) = 5x_{21}^2 + x_{21}x_{11}, f_{22}(\mathbf{x}) = x_{21}(x_{12} + x_{13}) + 10x_{22}^2, f_{23}(\mathbf{x}) = 15x_{23}^2, f_{24}(\mathbf{x}) = x_{24}^2, f_{25}(\mathbf{x}) = x_{25}^2 + x_{22}(x_{31} + x_{32} + x_{33} + x_{34})$, respectively. Furthermore, the objective functions for the agents in cluster 3 are respectively defined as $f_{31}(\mathbf{x}) = 2x_{31}^2 + x_{31}(x_{11} + x_{23}), f_{32}(\mathbf{x}) = x_{32}^2, f_{33}(\mathbf{x}) = x_{33}^2, f_{34}(\mathbf{x}) = x_{34}^2$.

The communication graphs for the agents in the 3 clusters are given in Fig. 2. In the following, we will consider the cases in which the Nash equilibrium is globally and locally characterized, successively.

4.1. Globally characterized Nash equilibrium

Let $\gamma_1 = 0$ in Example 1. Through direct calculation, it can be derived that the unique Nash equilibrium is $\mathbf{x}^* = \mathbf{0}_{12}$. Moreover, it can be easily verified that Assumption 3 is satisfied. With $\hat{\mathbf{x}}(0) = [-10, -8, -6, -4, 0, 2, 5, 7, 9, 10, 12, 8]^T, b_{ij} = 0.1, k_{ij} = 0.5$ and w_{ij} respectively chosen as 5, 6, 7, 8, 9, 10, 11, 12, 4, 3, 2, 1 for the 12 agents, the agents' estimated actions generated by the proposed method in (3) are given in Fig. 3. From Fig. 3, it can be seen that $\hat{\mathbf{x}}(t)$ converges to a small neighborhood of the Nash equilibrium though $\hat{\mathbf{x}}(0)$ is far away from the actual Nash equilibrium thus verifying Theorem 1.

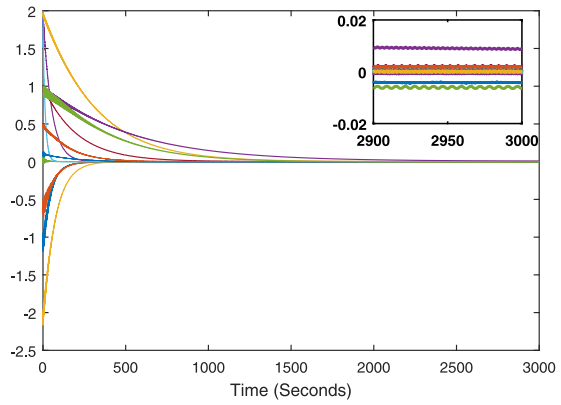


Fig. 4. The agents' estimated actions $\hat{x}_{ij}(t), i \in \{1,2,3\}, j \in \{1,2,\ldots,m_i\}$ generated by the proposed method in (3).

4.2. Locally characterized Nash equilibrium

To simulate the N-cluster noncooperative game in which the Nash equilibrium is locally characterized, we let $\gamma_1 = 1$ in Example 1. In this case, the stationary condition (i.e., the pseudo-gradient vector is equal to zero) gives two isolated points $\mathbf{x}' = [\frac{761}{240}, \frac{761}{1920}, 0, -\frac{2283}{6400}, \frac{761}{19200}, 0, 0, 0, -\frac{761}{960}, 0, 0, 0]^T$ and $\mathbf{x}'' = \mathbf{0}_{12}$. However, $\frac{\partial^2 f_1(\mathbf{x})}{\partial x_{11}^2} < 0$ for $\mathbf{x} = \mathbf{x}'$, which violates Assumption 4, while $\mathbf{x}''$ is a Nash equilibrium that satisfies Assumptions 4–5. Hence, $\mathbf{x}^* = \mathbf{x}''$ is exponentially stable by tuning the parameters δ, ϵ and b according to Theorem 2.

The agents' estimated actions generated by the proposed method in (3) are shown in Fig. 4 with $\hat{\mathbf{x}}(0) = [-1, -0.5, -2, 2, 0, 2, 1, 0.1, 0.5, 2, 1, 1]^T$ (the other variables are the same as the ones in Section 4.1). From the simulation result, it can be seen that $\hat{\mathbf{x}}(t)$ converges to a small neighborhood of the Nash equilibrium under the given conditions thus verifying Theorem 2.

It is worth noting that the frequencies chosen in the simulations violate the conditions required in Frihauf et al. (2012) and Menon and Baras (2014). However, the convergence can still be achieved by utilizing the proposed method, which is in line with Theorems 1–2. Moreover, we can directly choose distributed control gains b_{ij}, k_{ij}, w_{ij} defined in (3) instead of choosing $\epsilon, b, \delta, \bar{b}_{ij}, \bar{w}_{ij}, \bar{k}_{ij}$ separately. Roughly speaking, Theorems 1–2 suggest that b_{ij}, k_{ij}, w_{ij} should be chosen to be sufficiently small and w_{ij} for $i \in \mathcal{N}, j \in \{1,2,\ldots,m_i\}$ should be distinct from each other.

5. Conclusion

In this paper, an extremum seeker is developed to seek the Nash equilibrium of the N-cluster noncooperative games. A dynamic average consensus protocol is adopted in the seeking strategy design. Based on the real-time measurements of the output values of the agents' objective functions, the proposed extremum seeker does not require the explicit expressions of the agents' local objective functions. By Lyapunov stability analysis, local and nonlocal convergence results are developed successively under certain conditions.

Appendix A. Proof of Theorem 1

The system in (5) can be written as

$$\begin{aligned}\delta\frac{d\bar{g}_i}{d\tau} &= -R_i^T\mathcal{L}_iR_i\bar{g}_i + \delta R_i^T\left(\frac{\partial\mathcal{F}_i(\mathbf{x})}{\partial\mathbf{x}}\right)^T\frac{d\mathbf{x}}{d\tau}\\ \frac{d\hat{x}_{ij}}{d\tau} &= -\epsilon\bar{k}_{ij}\frac{f_i(\mathbf{x})}{m_i}b_{ij}\sin(\bar{w}_{ij}\tau)\\ &\quad + \epsilon\bar{k}_{ij}\left(f_i(\mathbf{x})/m_i - g_{ij}\right)b_{ij}\sin(\bar{w}_{ij}\tau).\end{aligned} \tag{14}$$

In the following, we follow the main ideas from averaging analysis to transform (14) into another system for the convenience of Lyapunov stability analysis.

Applying Taylor expansion to (14) gives

$$\begin{aligned}\frac{d\hat{x}_{ij}}{d\tau} &= -b\epsilon\frac{\bar{b}_{ij}\bar{k}_{ij}}{m_i}(f_i(\bar{\mathbf{x}}+\mathbf{x}^*) + b\frac{\partial f_i}{\partial\mathbf{x}}(\bar{\mathbf{x}}+\mathbf{x}^*)^T\times\\ &[\bar{b}_{ij}\sin(\bar{w}_{ij}\tau)]_{vec} + b^2H_i(\bar{\mathbf{x}}+\mathbf{x}^*,\tau))\sin(\bar{w}_{ij}\tau)\\ &+ \epsilon\bar{k}_{ij}\left(f_i(\mathbf{x})/m_i - g_{ij}\right)b_{ij}\sin(\bar{w}_{ij}\tau),\end{aligned} \tag{15}$$

$\forall i \in \mathcal{N}, j \in \{1,2,\ldots,m_i\}$ in which $b^2H_i(\bar{\mathbf{x}}+\mathbf{x}^*,\tau)$ represents for the residual terms of the Taylor expansion.

To facilitate the subsequent analysis, we follow Tan et al. (2006) and Ye and Hu (2016a) and Section 10.4 of Khalil (2002) to define a new coordinate as

$$\bar{x}_{ij}(\tau) = y_{ij}(\tau) + \epsilon b\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\xi_{ij}(\tau,\mathbf{y}), \tag{16}$$

in which $\mathbf{y} = [y_{ij}]_{vec}$, and

$$\begin{aligned}\xi_{ij}(\tau,\mathbf{y}) &= -\int_0^\tau\Big(f_i(\mathbf{y}+\mathbf{x}^*) + b\frac{\partial f_i}{\partial\mathbf{x}}(\mathbf{y}+\mathbf{x}^*)^T\times\\ &[\bar{b}_{ij}\sin(\bar{w}_{ij}\sigma)]_{vec}\Big)\sin(\bar{w}_{ij}\sigma) - \frac{b\bar{b}_{ij}}{2}\frac{\partial f_i}{\partial x_{ij}}(\mathbf{y}+\mathbf{x}^*)d\sigma.\end{aligned} \tag{17}$$

By choosing $\bar{w}_{ij} \neq \bar{w}_{kl}$ for distinct i, k or j, l, the integrand of (17) is periodic with zero mean for $\mathbf{y}$ that belongs to any nonempty compact set Ω that contains the origin. Recalling that $f_{ij}(\mathbf{x})$ is a smooth function according to Assumption 2, ξ_{ij} is periodic in τ and is bounded for $\mathbf{y} \in \Omega$. Moreover, $\frac{\partial\xi_{ij}}{\partial\tau} = -((f_i(\mathbf{y}+\mathbf{x}^*)+b\frac{\partial f_i}{\partial\mathbf{x}}(\mathbf{y}+\mathbf{x}^*)^T[\bar{b}_{ij}\sin(\bar{w}_{ij}\tau)]_{vec})\sin(\bar{w}_{ij}\tau) - \frac{b\bar{b}_{ij}}{2}\frac{\partial f_i}{\partial x_{ij}}(\mathbf{y}+\mathbf{x}^*))$ and $\frac{\partial\xi_{ij}}{\partial\mathbf{y}} = -\int_0^\tau(\frac{\partial f_1(\mathbf{y}+\mathbf{x}^*)}{\partial\mathbf{y}}+b\frac{\partial}{\partial\mathbf{y}}(\frac{\partial f_i}{\partial\mathbf{x}}(\mathbf{y}+\mathbf{x}^*)^T[\bar{b}_{ij}\sin(\bar{w}_{ij}\sigma)]_{vec}))\sin(\bar{w}_{ij}\sigma) - \frac{b\bar{b}_{ij}}{2}\frac{\partial}{\partial\mathbf{y}}(\frac{\partial f_i}{\partial x_{ij}}(\mathbf{y}+\mathbf{x}^*))d\sigma$ are periodic in τ and bounded for $\mathbf{y} \in \Omega$ by the smoothness of $f_{ij}(\mathbf{x})$ in Assumption 2.

Recalling that

$$\begin{aligned}\frac{d\bar{x}_{ij}}{d\tau} &= \frac{dy_{ij}}{d\tau} + \epsilon b\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\frac{d\xi_{ij}}{d\tau}\\ &= \frac{dy_{ij}}{d\tau} + \epsilon b\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}(\frac{\partial\xi_{ij}}{\partial\tau} + \left(\frac{\partial\xi_{ij}}{\partial\mathbf{y}}\right)^T\frac{d\mathbf{y}}{d\tau}),\end{aligned} \tag{18}$$

by (16), we have

$$\begin{aligned}\frac{d\mathbf{y}}{d\tau} &= \left(I + \epsilon b\mathrm{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\right\}\left(\frac{\partial\boldsymbol{\xi}}{\partial\mathbf{y}}\right)^T\right)^{-1}\\ &\times\left(-\epsilon b^2\mathrm{diag}\left\{\frac{\bar{b}_{ij}^2\bar{k}_{ij}}{2m_i}\right\}\mathcal{P}(\mathbf{y}+\mathbf{x}^*)\right.\\ &- \epsilon b\mathrm{diag}\left\{\frac{\bar{b}_{ij}\bar{k}_{ij}}{m_i}\right\}\Theta(\bar{\mathbf{x}},\mathbf{y},\tau)\\ &\left.+\epsilon b\mathrm{diag}\{\bar{k}_{ij}\bar{b}_{ij}\}\left[\left(\frac{f_i(\mathbf{x})}{m_i} - g_{ij}\right)\sin(\bar{w}_{ij}\tau)\right]_{vec}\right),\end{aligned} \tag{19}$$

where $\Theta(\bar{\mathbf{x}},\mathbf{y},\tau) = [\Theta_{ij}]_{vec}$, $\Theta_{ij} = ([(f_i(\bar{\mathbf{x}}+\mathbf{x}^*) + b\frac{\partial f_i}{\partial\mathbf{x}}(\bar{\mathbf{x}}+\mathbf{x}^*)^T[\bar{b}_{ij}\sin(\bar{w}_{ij}\tau)]_{vec})]\sin(\bar{w}_{ij}\tau) + b^2H_i(\bar{\mathbf{x}}+\mathbf{x}^*,\tau)\sin(\bar{w}_{ij}\tau)) - ((f_i(\mathbf{y}+\mathbf{x}^*)+b\frac{\partial f_i}{\partial\mathbf{x}}(\mathbf{y}+\mathbf{x}^*)^T[\bar{b}_{ij}\sin(\bar{w}_{ij}\tau)]_{vec}))\sin(\bar{w}_{ij}\tau)$ and $\boldsymbol{\xi} = [\xi_{ij}]_{vec}$. Note that (19) is derived by noticing that as $\frac{\partial\xi_{ij}}{\partial\mathbf{y}}$ for all $i \in \mathcal{N}, j \in \{1,2,\ldots,m_i\}$ are bounded, $I + \epsilon b\mathrm{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\right\}(\frac{\partial\boldsymbol{\xi}}{\partial\mathbf{y}})^T$ is nonsingular and $\left(I + \epsilon b\mathrm{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\right\}(\frac{\partial\boldsymbol{\xi}}{\partial\mathbf{y}})^T\right)^{-1} = I + O(\epsilon b)$ by choosing ϵb to be sufficient small for $\mathbf{y} \in \Omega$.

Now, we are ready to facilitate the Lyapunov stability analysis.

Noticing that the matrices $R_i^T\mathcal{L}_iR_i$ for all $i\in\mathcal{N}$ are symmetric positive definite under Assumption 1, there are symmetric positive definite matrices P_i, Q_i such that $(R_i^T\mathcal{L}_iR_i)P_i+P_i(R_i^T\mathcal{L}_iR_i)=Q_i$. Hence, we define the Lyapunov candidate function as

$$V=\frac{c}{2}\mathbf{y}^T\left(\text{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}^2}{2m_i}\right\}\right)^{-1}\mathbf{y}+(1-c)\sum_{i=1}^{N}\bar{\mathbf{g}}_i^TP_i\bar{\mathbf{g}}_i, \tag{20}$$

where $c\in(0,1)$ is a constant.

Then, for $\mathbf{E}_r=[\mathbf{y}^T,\bar{\mathbf{g}}^T]^T\in\Omega$, the time derivative of the Lyapunov candidate function along the given trajectory is

$$\begin{aligned}\frac{dV}{d\tau}=&c\mathbf{y}^T\left(\text{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}^2}{2m_i}\right\}\right)^{-1}\frac{d\mathbf{y}}{d\tau}\\&+(1-c)\sum_{i=1}^{N}\left(\frac{d\bar{\mathbf{g}}_i}{d\tau}\right)^TP_i\bar{\mathbf{g}}_i+(1-c)\sum_{i=1}^{N}\bar{\mathbf{g}}_i^TP_i\frac{d\bar{\mathbf{g}}_i}{d\tau}\\=&c\mathbf{y}^T\left(\text{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}^2}{2m_i}\right\}\right)^{-1}\left(I+\epsilon b\text{diag}\left\{\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\right\}\right.\\&\times\left.\left(\frac{\partial\boldsymbol{\xi}}{\partial\mathbf{y}}\right)^T\right)^{-1}\left(-\epsilon b^2\text{diag}\left\{\frac{\bar{b}_{ij}^2\bar{k}_{ij}}{2m_i}\right\}\mathcal{P}(\mathbf{y}+\mathbf{x}^*)\right.\\&-\epsilon b\text{diag}\left\{\frac{\bar{b}_{ij}\bar{k}_{ij}}{m_i}\right\}\Theta(\bar{\mathbf{x}},\mathbf{y},\tau)\\&+\epsilon b\text{diag}\{\bar{k}_{ij}\bar{b}_{ij}\}\left.\left[\left(\frac{f_i(\mathbf{x})}{m_i}-g_{ij}\right)\sin(\bar{w}_{ij}\tau)\right]_{vec}\right)\\&-(1-c)\sum_{i=1}^{N}\frac{\lambda_{min}(Q_i)}{\delta}\|\bar{\mathbf{g}}_i\|^2\\&+2(1-c)\sum_{i=1}^{N}\bar{\mathbf{g}}_i^TP_iR_i^T\left(\frac{\partial\mathcal{F}_i(\mathbf{x})}{\partial\mathbf{x}}\right)^T\frac{d\mathbf{x}}{d\tau}.\end{aligned} \tag{21}$$

If this is the case,

$$\begin{aligned}\frac{dV}{d\tau}\le&-\epsilon b^2c\mathbf{y}^T\mathcal{P}(\mathbf{y}+\mathbf{x}^*)\\&-(1-c)\sum_{i=1}^{N}\frac{\lambda_{min}(Q_i)}{\delta}\|\bar{\mathbf{g}}_i\|^2+\Xi(\mathbf{y},\bar{\mathbf{g}},\bar{\mathbf{x}},\tau),\end{aligned} \tag{22}$$

where

$$\begin{aligned}\Xi(\mathbf{y},\bar{\mathbf{g}},\bar{\mathbf{x}},\tau)=&c\mathbf{y}^T\left(\text{diag}\left\{(\bar{k}_{ij}\bar{b}_{ij}^2)/(2m_i)\right\}\right)^{-1}\times\\&\left(I+\epsilon b\text{diag}\left\{(\bar{k}_{ij}\bar{b}_{ij})/m_i\right\}((\partial\boldsymbol{\xi}/\partial\mathbf{y}))^T\right)^{-1}\times\\&\left(-\epsilon b^2\text{diag}\left\{(\bar{b}_{ij}^2\bar{k}_{ij})/(2m_i)\right\}\mathcal{P}(\mathbf{y}+\mathbf{x}^*)\right.\\&-\epsilon b\text{diag}\left\{(\bar{b}_{ij}\bar{k}_{ij})/m_i\right\}\Theta(\bar{\mathbf{x}},\mathbf{y},\tau)\\&\left.+\epsilon b\text{diag}\{\bar{k}_{ij}\bar{b}_{ij}\}\left[\left(f_i(\mathbf{x})/m_i-g_{ij}\right)\sin(\bar{w}_{ij}\tau)\right]_{vec}\right)\\&+2(1-c)\sum_{i=1}^{N}\bar{\mathbf{g}}_i^TP_iR_i^T\left(\partial\mathcal{F}_i(\mathbf{x})/\partial\mathbf{x}\right)^T(d\mathbf{x})/(d\tau)\\&+\epsilon b^2c\mathbf{y}^T\mathcal{P}(\mathbf{y}+\mathbf{x}^*).\end{aligned} \tag{23}$$

Then, for $\mathbf{E}_r\in\Omega$, there are positive constants α_1, α_2 and ν_1, ν_2 (depend on Ω) such that $\|\Xi(\mathbf{y},\bar{\mathbf{g}},\bar{\mathbf{x}},\tau)\|\le\epsilon b\alpha_1\|\mathbf{y}\|\|\bar{\mathbf{g}}\|+2(1-c)\|\bar{\mathbf{g}}\|(\nu_1(\epsilon b+b)+\epsilon b\alpha_2\|\bar{\mathbf{g}}\|)+\nu_2(\epsilon^2b^3+\epsilon^3b^3+\epsilon^3b^4+\epsilon^2b^4+\epsilon^2b^2+\epsilon b^3)$, by noticing that $\left(I+\epsilon b\text{diag}\{\frac{\bar{k}_{ij}\bar{b}_{ij}}{m_i}\}(\frac{\partial\boldsymbol{\xi}}{\partial\mathbf{y}})^T\right)^{-1}=I+O(\epsilon b)$ for sufficiently small ϵb and $f_i(\bar{\mathbf{x}}+\mathbf{x}^*)-f_i(\mathbf{y}+\mathbf{x}^*)=O(\epsilon b)$, $\frac{\partial f_i}{\partial\mathbf{x}}(\bar{\mathbf{x}}+\mathbf{x}^*)-\frac{\partial f_i}{\partial\mathbf{x}}(\mathbf{y}+\mathbf{x}^*)=O(\epsilon b)$, $H_i=O(1)$, $\left|\left|\frac{f_i(\mathbf{x})}{m_i}-g_{ij}\right|\right|\le\mu_{ij}\|\bar{\mathbf{g}}_i\|$ for some positive constant μ_{ij}, $\forall j\in\{1,2,\ldots,m_i\}$, $i\in\mathcal{N}$. Moreover, $\left|\left|\frac{d\bar{x}_{ij}}{d\tau}\right|\right|\le\left|\left|\frac{d\hat{x}_{ij}}{d\tau}\right|\right|+\left|\left|b\bar{b}_{ij}\bar{w}_{ij}\cos(\bar{w}_{ij}\tau)\right|\right|\le\left|\left|\epsilon\bar{k}_{ij}\frac{b_{ij}f_i(\mathbf{x})}{m_i}\sin(\bar{w}_{ij}\tau)\right|\right|+\|\epsilon\bar{k}_{ij}\left(\frac{f_i(\mathbf{x})}{m_i}-g_{ij}\right)b_{ij}\sin(\bar{w}_{ij}\tau)\|+\left|\left|b\bar{b}_{ij}\bar{w}_{ij}\cos(\bar{w}_{ij}\tau)\right|\right|$, where $f_i(\mathbf{x})$ for all $i\in\mathcal{N}$ are bounded for $\mathbf{E}_r(\tau)\in\Omega$.

Note that $(\epsilon b+b)\|\bar{\mathbf{g}}\|\le\frac{\epsilon^2b^2}{2\theta_1}+\frac{\theta_1\|\bar{\mathbf{g}}\|^2}{2}+\frac{b^2}{2\theta_2}+\frac{\theta_2\|\bar{\mathbf{g}}\|^2}{2}$ where θ_1,θ_2 are positive constants that can be arbitrarily chosen. Moreover, $\epsilon b\|\mathbf{y}\|\|\bar{\mathbf{g}}\|\le\frac{\epsilon^2b^2\|\mathbf{y}\|^2}{2\theta_3}+\frac{\theta_3\|\bar{\mathbf{g}}\|^2}{2}$, where θ_3 is a positive constant that can be arbitrarily chosen. Hence, for $\mathbf{E}_r(\tau)\in\Omega$,

$$\begin{aligned}\frac{dV}{d\tau}\le&-\epsilon b^2c\mathbf{y}^T\mathcal{P}(\mathbf{y}+\mathbf{x}^*)-(1-c)\sum_{i=1}^{N}\frac{\lambda_{min}(Q_i)}{\delta}\|\bar{\mathbf{g}}_i\|^2\\&+2(1-c)\nu_1\left(\frac{\epsilon^2b^2}{2\theta_1}+\frac{\theta_1\|\bar{\mathbf{g}}\|^2}{2}+\frac{b^2}{2\theta_2}+\frac{\theta_2\|\bar{\mathbf{g}}\|^2}{2}\right)\\&+\alpha_1\left(\frac{\epsilon^2b^2\|\mathbf{y}\|^2}{2\theta_3}+\frac{\theta_3\|\bar{\mathbf{g}}\|^2}{2}\right)+2\epsilon b(1-c)\alpha_2\|\bar{\mathbf{g}}\|^2\\&+\nu_2(\epsilon^2b^3+\epsilon^3b^3+\epsilon^3b^4+\epsilon^2b^4+\epsilon^2b^2+\epsilon b^3)\\\le&-\frac{\epsilon b^2c}{2}W(\mathbf{y})-\left(\frac{(1-c)\underline{\lambda}}{\delta}-\frac{\epsilon b^2c}{2}-\frac{\alpha_1\theta_3}{2}\right.\\&\left.-(1-c)\nu_1\theta_1-\theta_2(1-c)\nu_1-2\epsilon b(1-c)\alpha_2\right)\|\bar{\mathbf{g}}\|^2\\&-\frac{\epsilon b^2c}{2}\|\bar{\mathbf{g}}\|^2-\frac{\epsilon b^2c}{2}W(\mathbf{y})\\&+\nu_2(\epsilon^2b^3+\epsilon^3b^3+\epsilon^3b^4+\epsilon^2b^4+\epsilon^2b^2+\epsilon b^3)\\&+\frac{\alpha_1\epsilon^2b^2}{2\theta_3}\|\mathbf{y}\|^2+\frac{(1-c)\nu_1\epsilon^2b^2}{\theta_1}+\frac{b^2(1-c)\nu_1}{\theta_2},\end{aligned} \tag{24}$$

where $\underline{\lambda}=\min\{\lambda_{min}(Q_i)\}$, for $i\in\mathcal{N}$ and $W(\mathbf{y})=\mathbf{y}^T\mathcal{P}(\mathbf{y}+\mathbf{x}^*)$ is a positive definite function by utilizing Assumption 3.

Therefore,

$$\begin{aligned}\frac{dV}{d\tau}\le&-\frac{\epsilon b^2c}{2}W(\mathbf{y})-\left(\frac{(1-c)\underline{\lambda}}{\delta}-\frac{\epsilon b^2c}{2}-\frac{\alpha_1\theta_3}{2}\right.\\&\left.-(1-c)\nu_1\theta_1-\theta_2(1-c)\nu_1-2\epsilon b(1-c)\alpha_2\right)\|\bar{\mathbf{g}}\|^2,\end{aligned} \tag{25}$$

given that $\frac{\epsilon b^2c}{2}\|\bar{\mathbf{g}}\|^2+\frac{\epsilon b^2c}{2}W(\mathbf{y})\ge\nu_2(\epsilon^2b^3+\epsilon^3b^3+\epsilon^3b^4+\epsilon^2b^4+\epsilon^2b^2+\epsilon b^3)+\frac{\alpha_1\epsilon^2b^2}{2\theta_3}\alpha_3+\frac{(1-c)\nu_1\epsilon^2b^2}{\theta_1}+\frac{b^2(1-c)\nu_1}{\theta_2}$, where α_3 is a positive constant such that $\|\mathbf{y}\|\le\alpha_3$ for $\mathbf{E}_r(\tau)\in\Omega$. Let $\beta_1\in\mathcal{K}$ be a function that satisfies $\beta_1(\|\mathbf{E}_r(\tau)\|)\le\|\bar{\mathbf{g}}\|^2+W(\mathbf{y})$.

Then, if

$$\|\mathbf{E}_r(\tau)\|\ge\beta_1^{-1}\left(\Phi(\epsilon,b)\right), \tag{26}$$

where $\Phi(\epsilon,b)=\frac{2\nu_2}{c}(\epsilon b+\epsilon^2b+\epsilon^2b^2+\epsilon b^2+\epsilon+b)+\frac{\alpha_1\epsilon\alpha_3}{c\theta_3}+\frac{2(1-c)\nu_1\epsilon}{\theta_1c}+\frac{2(1-c)\nu_1}{\theta_2\epsilon c}$, we have,

$$\frac{dV}{d\tau}\le-\frac{\epsilon b^2c}{2}\beta_1(\|\mathbf{E}_r\|), \tag{27}$$

by choosing $\delta<\delta^*$ where

$$\delta^*=\frac{2(1-c)\underline{\lambda}}{2\epsilon b^2c+4\epsilon b(1-c)\alpha_2+2(\theta_1+\theta_2)(1-c)\nu_1+\alpha_1\theta_3}.$$

Note that $\Phi(\epsilon,b)$ can be made arbitrarily small by choosing ϵ,b to be sufficiently small and with fixed ϵ,b, choose θ_2 to be sufficiently large. Moreover, with fixed ϵ,b and θ_2, choose $\delta<\delta^*$.

Moreover, by the definition of the Lyapunov candidate function, it can be derived that there are positive constants M_1,M_2

such that

$$M_1\|\mathbf{E}_r\|^2 \le V(\mathbf{E}_r) \le M_2\|\mathbf{E}_r\|^2. \tag{28}$$

The rest of the proof follows the proof of Theorem 4.18 in Khalil (2002) to take any $r > 0$ such that $\mathbf{B}_r \subset \Omega$ and choose ϵ and b to be sufficiently small such that $\mu < \sqrt{\frac{M_1}{M_2}}r$, where $\mu = \beta_1^{-1}(\Phi(\epsilon, b))$. Let $\rho = M_1 r^2$, $\eta = M_2\mu^2$ and define $\Omega_\eta = \{\mathbf{E}_r \in \mathbf{B}_r | V(\mathbf{E}_r) \le \eta\}$, $\Omega_\rho = \{\mathbf{E}_r \in \mathbf{B}_r | V(\mathbf{E}_r) \le \rho\}$. Then, $\mathbf{B}_\mu \subset \Omega_\eta \subset \Omega_\rho \subset \mathbf{B}_r \subset \Omega$. Note that a solution starting from Ω_η or Ω_ρ cannot leave it as the time-derivative of the Lyapunov candidate function is negative on the boundary.

Suppose that $\Delta = \sqrt{\frac{M_1}{M_2}}r$, i.e., $\|\mathbf{E}_r(\tau_0)\| \le \sqrt{\frac{M_1}{M_2}}r$. Then, $\mathbf{E}_r(\tau_0) \in \Omega_\rho$. If this is the case, a solution starting inside Ω_η satisfies

$$\|\mathbf{E}_r(\tau)\| \le \sqrt{\frac{M_2}{M_1}}\beta_1^{-1}(\Phi(\epsilon, b)), \tag{29}$$

for all $\tau \ge \tau_0$ as $\Omega_\eta \subset \left\{\mathbf{E}_r | \|\mathbf{E}_r\| \le \sqrt{\frac{M_2}{M_1}}\mu\right\}$. Moreover, if $\mathbf{E}_r(\tau_0)$ is outside Ω_η (but inside Ω_ρ), it must enter Ω_η in finite time. Denote the first time it enters Ω_η as $\mathcal{T}$. Then, for all $\tau \in [\tau_0, \tau_0 + \mathcal{T}]$,

$$\frac{dV}{d\tau} \le -\frac{\epsilon b^2 c}{2}\beta_1(\|\mathbf{E}_r\|) \le -\frac{\epsilon b^2 c}{2}\beta_1\left(\sqrt{\frac{V}{M_2}}\right). \tag{30}$$

Noticing that r can be arbitrarily large as Ω can be any compact set that contains the origin, the result can be derived with $v = \sqrt{\frac{M_2}{M_1}}\beta_1^{-1}(\Phi(\epsilon, b)) + O(\epsilon b)$ by further utilizing Lemmas 3.4 and 4.4 in Khalil (2002) and (16).

Appendix B. Proof of Theorem 2

The theorem can be proved by the following steps:
Step 1: Follow the proof of Theorem 1 to obtain (19).
Step 2: Show that by Assumptions 4–5, there exists a positive definite function $V_1(\mathbf{y}) : D_0 \to \mathbb{R}$, $D_0 = \{\mathbf{y} \in \mathbb{R}^{\sum_{i=1}^N m_i} | \ \|\mathbf{y}\| \le r_0\}$ for some positive constant r_0, such that $c_1\|\mathbf{y}\|^2 \le V_1(\mathbf{y}) \le c_2\|\mathbf{y}\|^2$, $-\left(\frac{\partial V_1}{\partial \mathbf{y}}\right)^T \text{diag}\left\{\frac{\bar{b}_{ij}^2\bar{k}_{ij}}{2m_i}\right\}\mathcal{P}(\mathbf{y} + \mathbf{x}^*) \le -c_3\|\mathbf{y}\|^2$, $\left\|\frac{\partial V_1}{\partial \mathbf{y}}\right\| \le c_4\|\mathbf{y}\|$, where c_1, c_2, c_3, c_4 are positive constants (Frihauf et al., 2012; Ye & Hu, 2017b; Ye et al., 2018).
Step 3: Define the Lyapunov candidate function as

$$V = cV_1(\mathbf{y}) + (1 - c)\sum_{i=1}^N \bar{g}_i^T P_i \bar{g}_i, \tag{31}$$

where $c \in (0, 1)$ is a constant, to show the stability of the equilibrium.

Note that the proof is similar to that of Theorem 1 and the detailed analysis is omitted due to space limitation.

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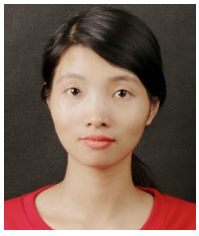
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